Outliers in Financial Time Series

Outlier detection: Consideration of extremes:

- *what to trust, what to model as risk, signal, other…*

- The same extreme value (return) can be a data error, a microstructure artifact, or the start of a new regime
- Many outlier detection methods exist, but method matters less than the decision you take in modelling
- Potential impacts:
  - Tails drive drawdowns(worst peak-to-through losses in return) - **small number of extreme days**
    - can be used to explain most of the drawdown
    - can be used to explain a large chunk of long-run performance
  - Model stability (avoid overfitting to a few points)
  - Regime detection (clusters of extremes)
  - Alerts / early warning (credit spread blowouts, liquidity events)
  - Overstate Sharpe ratio
  - Other…

Outliers Handling

Outliers shape the empirical distribution (especially tails) — so treatment affects both risk estimates and model behaviour.

Treat outliers differently for:

1) Explanatory variables (X)

- **If erroneous / contaminated - fix / impute / replace (data-quality) – DON'T Drop Dates**
- **Clip / winsorize/** replace feature values with clip(X, q01, q99) (or robust z-score cap)
- **Robust scaling** - robust scaling + robust loss / regularization
- **Rank transforms (replace magnitudes with order)**
- **Encode stress: is_outlier_X, outlier_score_X, rolling outlier density (regime signal)**

2) Target variable (y)

- **If erroneous → correct / drop / replace** (data-quality)
- **If genuine market move → keep tails, model them explicitly**
- Heavy-tailed likelihood (Student-t; or mixture/jump)
- Regime modelling (normal vs stress; often gated by X/stress score)
- Episode indicators (shock-day flags, stress-cluster counts)

Key caution

- Don't let rare points hijack training (robust losses/transformations help).
- Define outliers with **rolling/expanding baselines** to avoid **look-ahead bias**.

Example – Univariate Time Series Outlier (SPY)

Example – Outlier Detection – x3 Methods

Method 1 - Global Z-score outliers (mean/SD standardization)

Let $\{x_t\}_{t=1}^{T}$ be returns.

Sample mean and standard deviation:

$$\bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t, \qquad s = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^2}$$

Z-score:

$$z_t = \frac{x_t - \bar{x}}{s}$$

$$\text{outlier at } t \iff |z_t| > \tau$$

$\tau = 3.$ standard deviations

Note 1: few extremes can inflate **s** and mask other outliers
heavy-tailed returns violate Gaussian intuition

Method 2 - Modified Z-score using MAD (robust standardization)

Robust location:

$$m = \text{median}(x_1, ..., x_T)$$

Median Absolute Deviation (MAD):

$$\text{MAD} = \text{median}\left(|x_t - m|\right)$$

$$\text{outlier at } t \iff |M_t| > \tau$$

$\tau = 3.5$ standard deviations

Note 2: Replace estimators (mean/SD) with **robust** ones (median/MAD)

Example – Outlier Detection – x3 Methods

Method 3 - Rolling-window Z-score (local mean/vol normalization)

- Financial returns are often **nonstationary in volatility (heteroskedastic)** and sometimes in mean
- Instead of a single global location/scale, estimate **time-varying local mean and local volatility** over a moving window, then flag points with unusually large *local* standardized residuals.
- This is a simple, nonparametric cousin of volatility models (e.g., local estimation vs GARCH).

Rolling (trailing) mean and standard deviation:

$$\mu_t = \frac{1}{w}\sum_{i=0}^{w-1} x_{t-i}, \qquad \sigma_t = \sqrt{\frac{1}{w-1}\sum_{i=0}^{w-1}(x_{t-i}-\mu_t)^2}$$

(defined for $t \geq w$)

Rolling z-score:

$$z_t^{(roll)} = \frac{x_t - \mu_t}{\sigma_t}$$

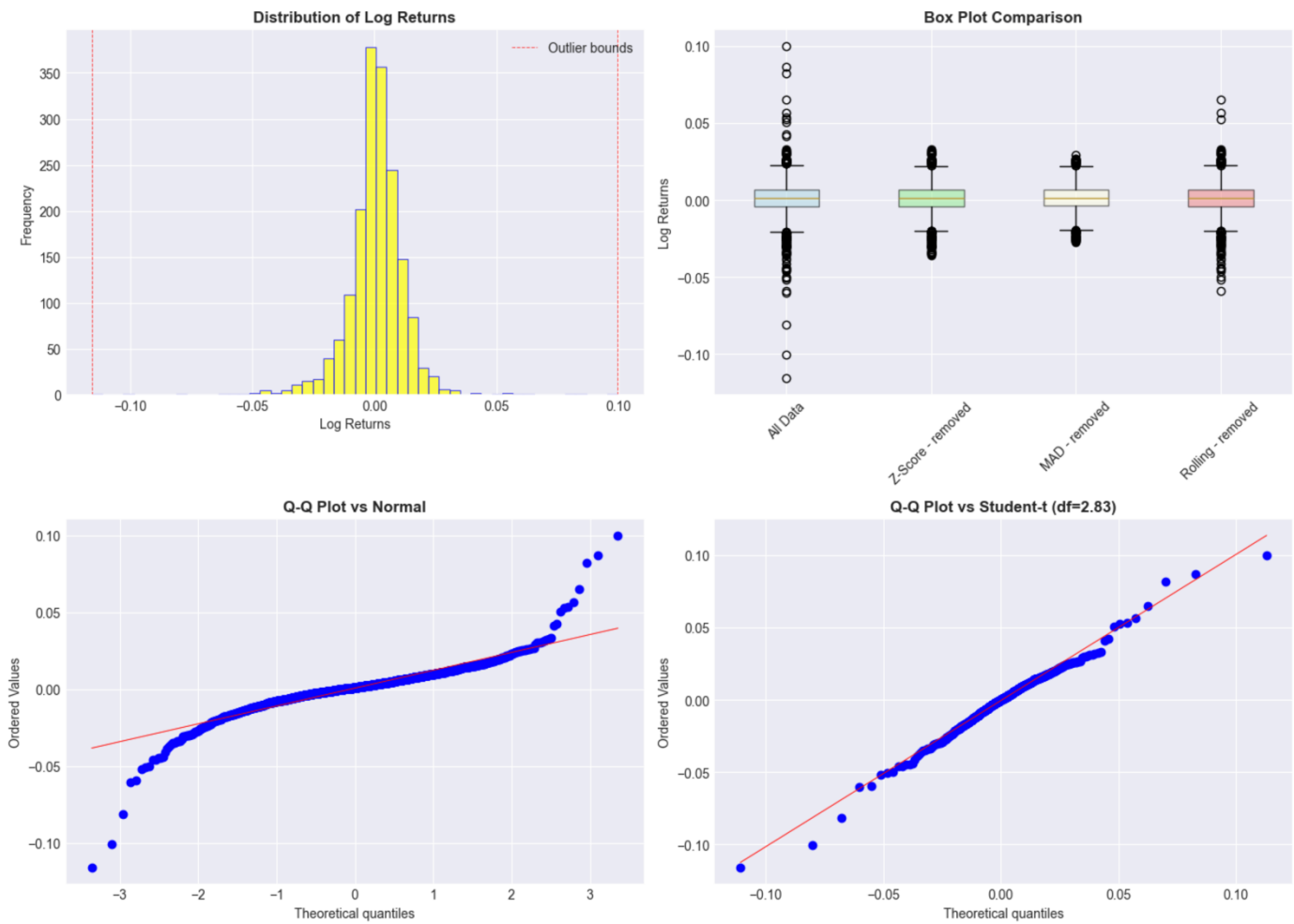
$\boldsymbol{\omega}$ – set to 84 $\qquad$ $\boldsymbol{\omega}$ – Options $252, 126, 84, 63$ days

$$\text{outlier at } t \iff \left|z_t^{(roll)}\right| > \tau$$

$$\tau = 3.$$

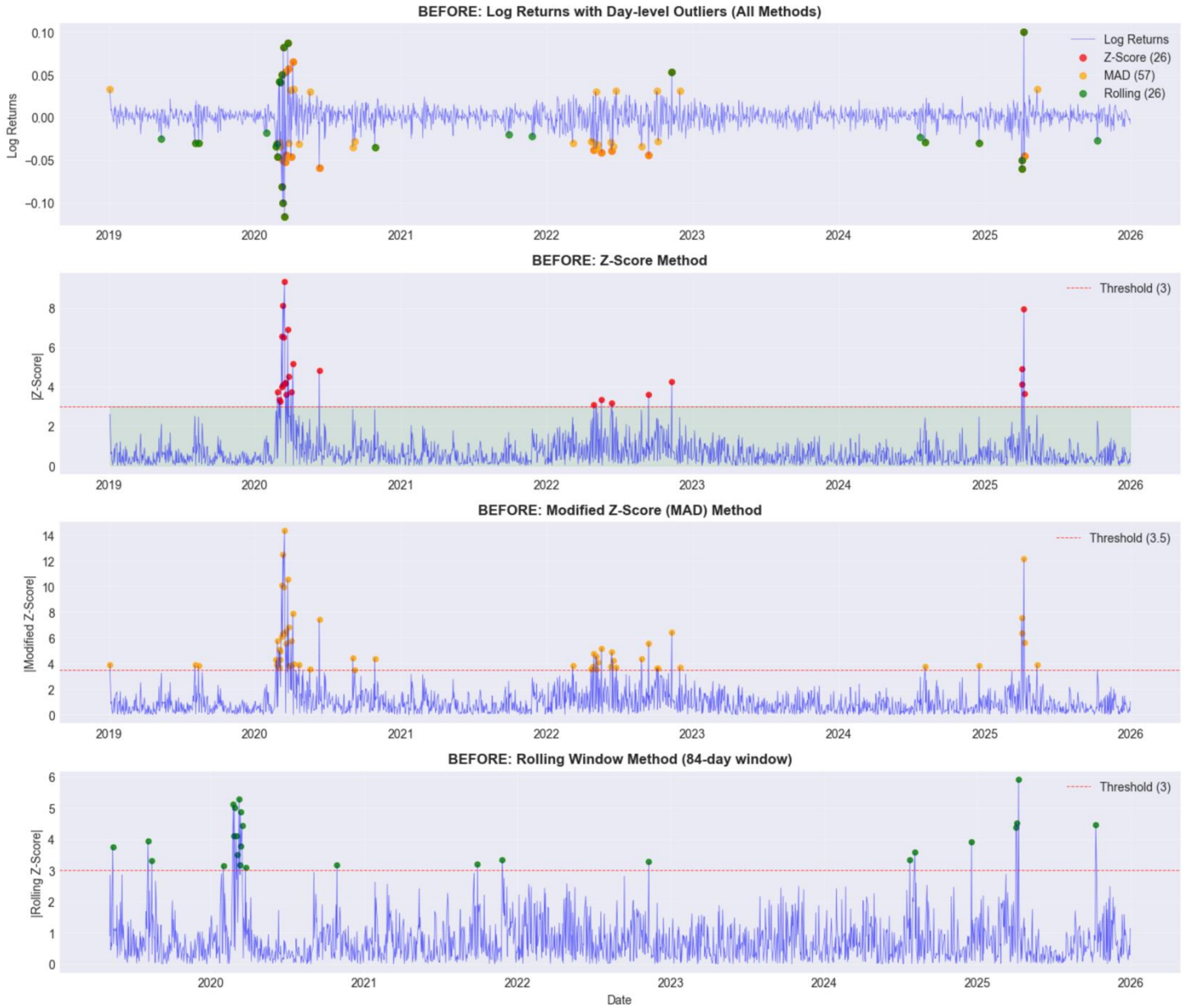
Note 3: "extremes" can be **normalized away** right after regime shifts

Example – Outlier Profiling Summary

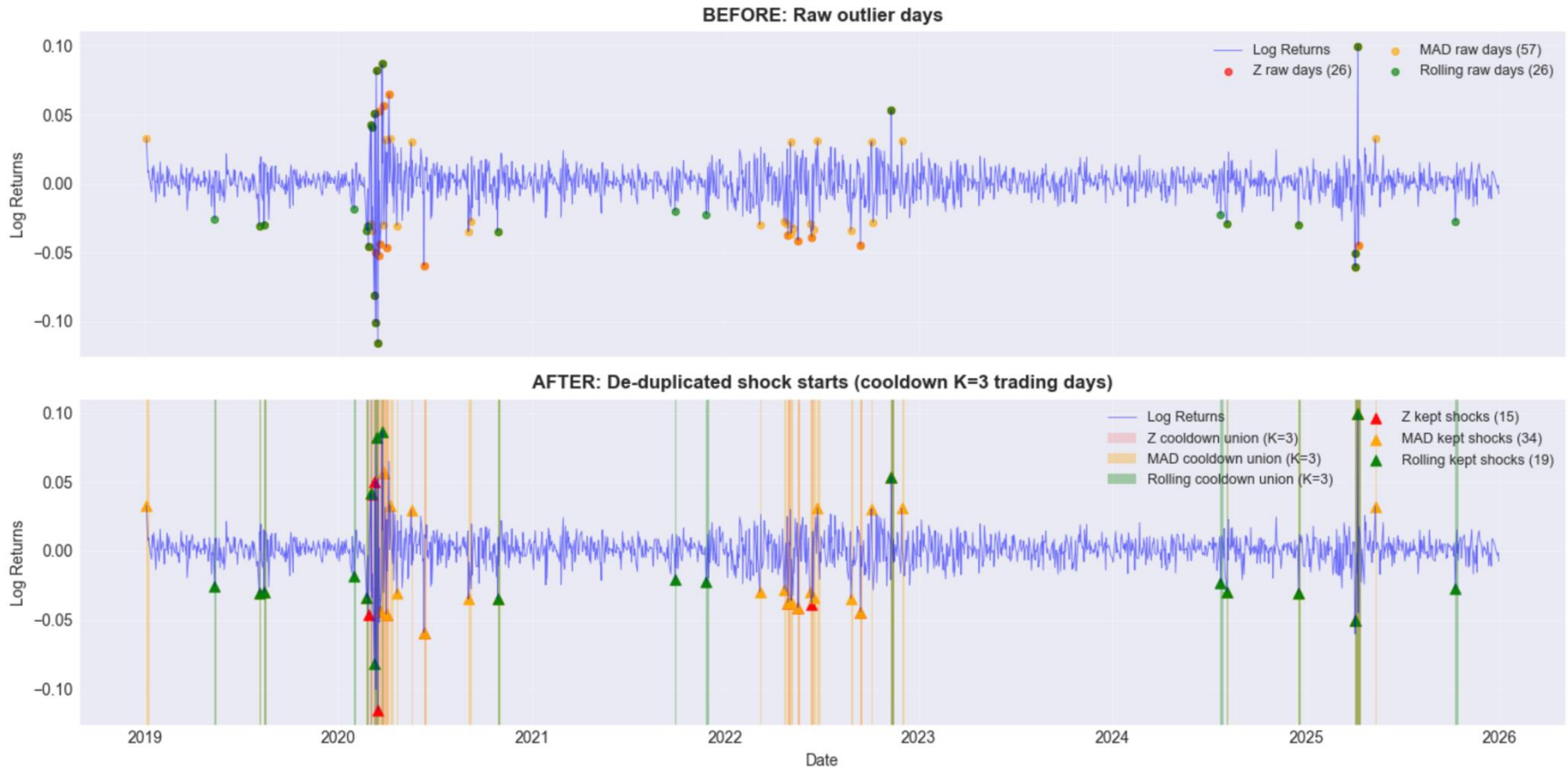


- **All three** outlier removal methods **shrink the tails**
- **Global Z-score removal** typically trims based on a single σ
- **MAD removal** is robust, it tends to treat a few extremes as extremes without being pulled by them
- **Rolling removal** adapts to volatility clustering
- **QQ plots** support each other

Example – Pointwise outliers (or single-day outliers)

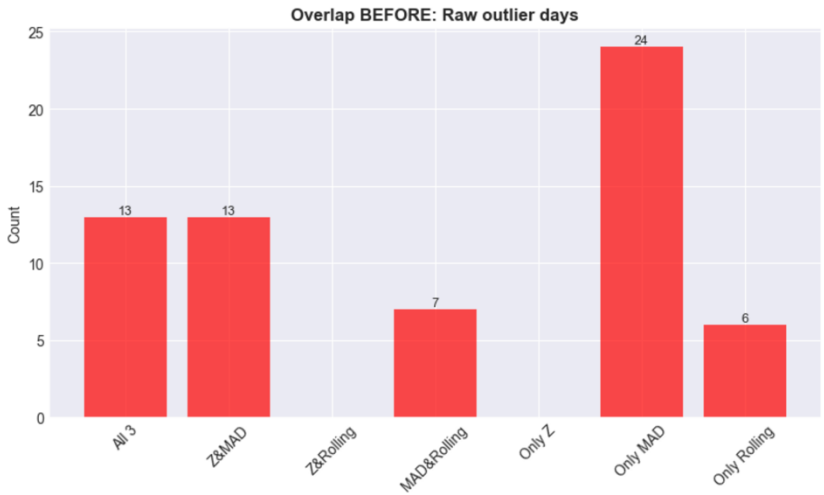


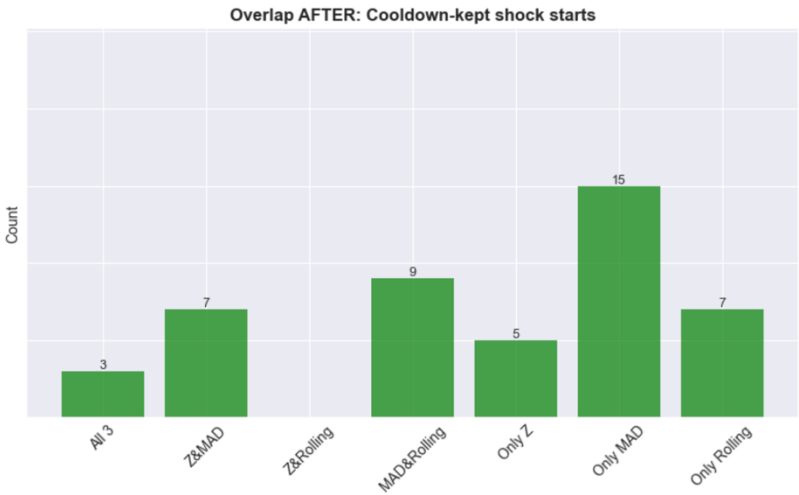
Example – Sustained outlier episodes (or run-based episodes)



- **De-duplicated shock events** (or **cooldown-filtered shocks**)
- **3 Day Window = Episode**

Example – Overlap Comparison: Day-Level vs De-duplicated Shocks





TOP 10 EXTREME EPISODES & METHODS

	Date	Log-Return	Detection Method
1.	2020-03-16:	-0.1159	(Detected by: Z,MAD,Roll)
2.	2020-03-12:	-0.1006	(Detected by: Z,MAD,Roll)
3.	2025-04-09:	0.0999	(Detected by: Z,MAD,Roll)
4.	2020-03-24:	0.0867	(Detected by: Z,MAD,Roll)
5.	2020-03-13:	0.0820	(Detected by: Z,MAD,Roll)
6.	2020-03-09:	-0.0813	(Detected by: Z,MAD,Roll)
7.	2020-04-06:	0.0650	(Detected by: Z,MAD)
8.	2025-04-04:	-0.0603	(Detected by: Z,MAD,Roll)
9.	2020-06-11:	-0.0594	(Detected by: Z,MAD)
10.	2020-03-26:	0.0567	(Detected by: Z,MAD)

In a Nutshell – Feasible Considerations 1

Don't "find the best outlier model"; Consider what to do with them

`Core:

- keep all dates
- reduce few outliers "hijacking" model training
- preserve stress information

For ***X*** (*features*) Practical Treatments:

- rolling winsorize/clip + outlier flag
- robust scaling (median/IQR) + outlier flag
- rank transform (percentile) + (optionally) keep raw too

Don't remove outliers; create two views of *X*

Make two versions of the same features:

1. X_clean = clipped / robust-scaled / rank-transformed (stable for learning)
2. X_raw = original values (kept for diagnostics and for stress flags)

Create stress indicators from raw features:

1. is_outlier_X (per feature)
2. stress_density (rolling % of outlier days)
3. "relationship break" residual z-scores (multivariate but simple)

In a Nutshell – Feasible Considerations 2

Modelling Choices: one model with stress features or two model regime

X (explanatory variables)

Option A (simplest, common): one e.g. XGBoost model

- Train on all dates
- Inputs = X_clean + stress flags (is_outlier_lag1, is_outlier_last5days)
- The model learns different behavior in stress without separate models

Option B two-model regime approach

- Train a normal model mainly on normal periods (using X_clean)
- Train a stress model on stress periods (again using X_clean + stress flags)

y (target)

- Fix only true errors
- Keep tail events

Important consideration

Cleaning outlier consideration:

- It is possible to remove or cap ("winsorize") outliers **only when we are confident they're bad data /**, not genuine tail events otherwise you erase real risk

Modelling consideration:

- If outliers are real, keep them and treat them explicitly: **heavy-tail likelihood (Student-t)**, **stress flags**, **sample weights**, or **regime labels / separate normal–stress model**

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Multivariate Outlier Detection

- **Multivariate outlier detection** engages with joint behaviour across many variables (often with **lags/lead–lag effects**)

- This is a **temporal dependency problem** (cross-asset interactions + time alignment), so it needs special treatment.